

IMPORTANT

Please use **Jupyter** notebook *power_plants_task.ipynb* to complete the assignment. Do not use separate .py files — the notebook should be entirely self-contained.

Make sure the notebook runs from start to finish without any errors.

To complete the assignment, you will need to **pip install pandas** library.

Please provide documentation that explains your solution. You may use an LLM to assist with the code, but make sure you describe the approach in your own words.

Please upload all your work to public **GitHub** repository and share the link with us.

TASK

There are three .csv files that we want to upload sequentially to the database: **gas_fr_plants.csv**, **gas_plants.csv**, and **wind_plants.csv**. For the sake of simplicity, the database is represented by another .csv file **database.csv**. These three files contain volume data for certain power plants.

Please analyse these three plant files to check for missing data or other errors. Pease show your working and thought process when dealing with this.

The goal is to complete **PowerPlants** class with several functions:

- **analyse_plant_data** : analyses and cleans data from a ***plants.csv** file, handling missing values, duplicates and invalid entries, returning a cleaned DataFrame
- **load_new_data_from_file** : loads and processes a cleaned DataFrame ready for saving
- **save_new_data** : saves loaded data to the **database.csv** file
- **get_data_from_database** : returns the most recently updated data for every plant in **database.csv**
- **aggregate_data_to_quarterly** : returns the quarterly mean, median and standard deviation for each wind or Gas plant in **database.csv** (Jan-Mar, Apr-Jun, Jul-Sep, Oct-Dec)
- **aggregate_data_to_country** : returns the country total power production by technology type in **database.csv**

```
In [ ]: class PowerPlants(object):
        def __init__(self):
            self.database_file = 'database.csv'

        def analyse_plant_data(self, file_path: str):
            pass

        def load_new_data_from_file(self, input_data: pd.DataFrame):
            pass

        def save_new_data(self, input_data: pd.DataFrame):
            pass

        def get_data_from_database(self):
            pass

        def aggregate_data_to_quarterly(self):
            pass

        def aggregate_data_to_country(self):
            pass
```

```
In [ ]: pp = PowerPlants()
new_data = pp.load_new_data_from_file(pp.analyse_plant_data('wind_plants.csv'))
pp.save_new_data(new_data)
new_data = pp.load_new_data_from_file(pp.analyse_plant_data('gas_plants.csv'))
pp.save_new_data(new_data)
new_data = pp.load_new_data_from_file(pp.analyse_plant_data('gas__fr_plants.csv'))
pp.save_new_data(new_data)
```

Please note:

- *Country* column from **plants.csv** files are stored as full country names in **database.csv**
- There maybe some missing data that needs to be populated (please assume 0 for this)
- The most recent update for each symbol is returned based on the **updatetime** column

Please see the below for the structure of expected results.

```
In [1]: from IPython.display import Image
Image('get_data_from_database.png')
```

Out[1]:

	date	country	SiteName	Technology	updatedby	update	time	Volume
	2024-01-01	France	Blenod-5	Gas	petroineos	2025-05-29 11:06:00.802833		6753.000000
	2024-01-02	France	Blenod-5	Gas	petroineos	2025-05-29 11:06:00.802833		3896.000000
	2024-01-03	France	Blenod-5	Gas	petroineos	2025-05-29 11:06:00.802833		3636.000000
	2024-01-04	France	Blenod-5	Gas	petroineos	2025-05-29 11:06:00.802833		5138.000000
	2024-01-05	France	Blenod-5	Gas	petroineos	2025-05-29 11:06:00.802833		5265.000000
	...	...	...	...	...	...		...
	2025-04-21	Great Britain	Hornsea-2	Wind	petroineos	2025-05-29 11:06:01.001653		711.231619
	2025-04-22	Great Britain	Hornsea-2	Wind	petroineos	2025-05-29 11:06:01.001653		808.534585
	2025-04-23	Great Britain	Hornsea-2	Wind	petroineos	2025-05-29 11:06:01.001653		142.450340
	2025-04-24	Great Britain	Hornsea-2	Wind	petroineos	2025-05-29 11:06:01.001653		392.082184
	2025-04-25	Great Britain	Hornsea-2	Wind	petroineos	2025-05-29 11:06:01.001653		920.847877

In [6]:

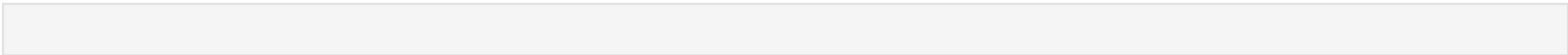
```
from IPython.display import Image
Image('month_average_min_mix.png')
```

#Note this image shows monthly with min/max yours will be aggregated to quarters with sd/median

Out[6]:

	Blenod-5 Mean	Blenod-5 Min	Blenod-5 Max	Pembroke-1 Mean	Pembroke-1 Min	Pembroke-1 Max	Pembroke-2 Mean	Pembroke-2 Min	Pembroke-2 Max	Hornsea-1 Mean	Hornsea-1 Min	Hornsea-1 Max	Hornsea-2 Mean	Hornsea-2 Min	Hornsea-2 Max
date															
2024-01-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2024-02-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2024-03-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2024-04-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2024-05-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2024-06-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2024-07-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2024-08-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2024-09-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2024-10-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2024-11-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2024-12-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2025-01-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2025-02-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2025-03-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2025-04-01	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

In []:



In [3]:

```
from IPython.display import Image
Image('country_sum.png')
```

Out[3]:

	country	Technology	Volume
	France	Gas	TBD
	Great Britain	Gas	TBD
		Wind	TBD

Good luck!!!

In []:

